



Fiona Stanley Fremantle Hospitals Group

OFFICIAL

Policy & Procedure

Electroconvulsive Therapy Registrar Role

Reference #: FSFHG-MENH-POL-0015

Scope

Site	Service/Department/Unit	Disciplines
Fremantle Hospital	Mental Health Service	Medical & Nursing

1. Introduction

Electroconvulsive Therapy (ECT) is a safe and effective treatment for a variety of psychiatric conditions. The administration of ECT involves a multidisciplinary team, with duties and responsibilities delegated throughout the team according to qualifications and seniority. This procedure outlines the role of the Psychiatric Registrar in ECT and applies to all Registrars involved in the delivery of ECT in alignment with the standards set out by the Chief Psychiatrist

2. Terminology

Electroconvulsive Therapy (ECT)	The application of electric current to specific areas of the head to produce a generalised seizure which is modified by general anaesthetic and the administration of a muscle relaxing agent. It is mainly used to treat severe depression, but is also indicated for patients with other conditions, including bipolar disorder, schizophrenia, schizoaffective disorder, catatonia, and neuroleptic malignant syndrome.
Impedance	The measure of the opposition that a circuit presents to a current when voltage is applied.
Thymapad	Single use treatment electrode

Valid ECT consent	ECT consent is valid when: • an ECT consent form is completed for voluntary patients • there are <7 days from obtaining consent to commencing treatment • the patient is informed that in consenting to a course of ECT they can withdraw consent at any time during treatment. Patients and carers are made aware that consents are usually for a course of treatments and the number of treatments is to be specified • a separate anaesthetic consent is completed • there is documentation of Mental Health Tribunal decision of approval available in the medical record for a patient receiving treatment under the MHA 2014 • when it is renewed after 3 months or when the approved/prescribed number of ECT treatments is completed whichever comes first. • consent for less than 18 yo may only be given with approval by the Mental Health Tribunal
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3. Policy

- The treating Consultant Psychiatrist is ultimately responsible for the decision to prescribe and administer ECT.
- This decision will be made in conjunction with the treating team responsible for the care of the patient and an Anaesthetist.
- The Anaesthetist is responsible for determining the physical fitness of a patient to undergo general anaesthesia

4. Procedure

4.1. Applying the Electroencephalogram (EEG) and ECT electrodes

- Perform hand hygiene
- Check and confirm the ECT prescription is documented on the Integrated ECT/Anaesthetic Form and the ECT consent is valid
- Confirm ECT sign in is completed
- Determine ECT electrode placement required and apply EEG leads accordingly

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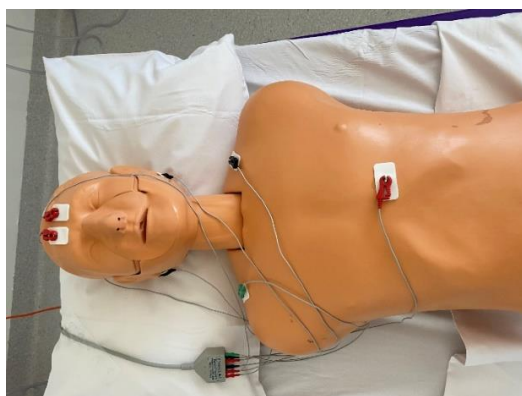


Figure 1: Bifrontal

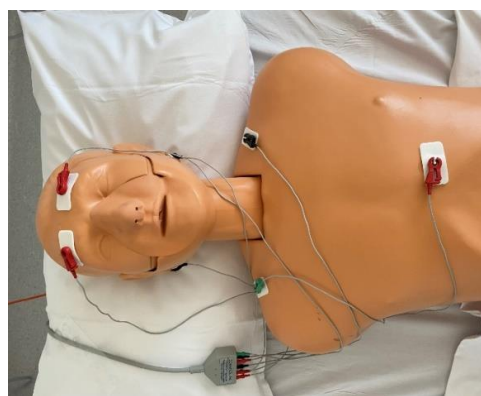


Figure 2: Bitemporal/Right unilateral

- Prepare the area for ECT electrode placement by wiping with a Reynard prep pad and drying with gauze

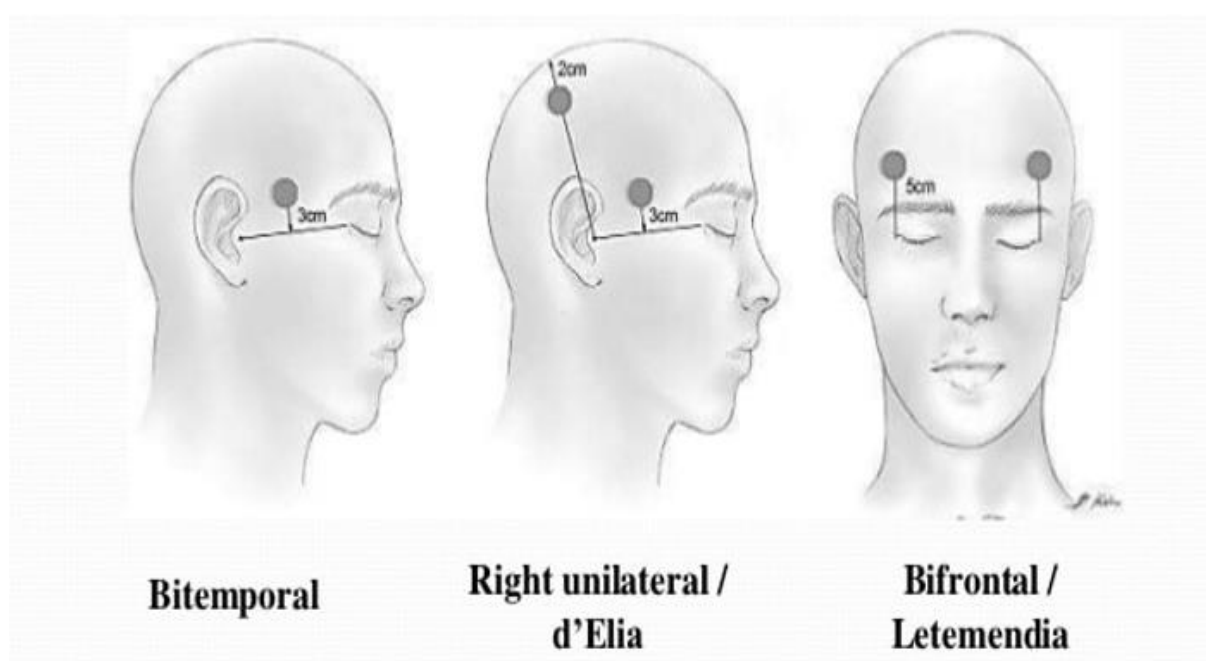


Figure 3: ECT electrode placement

Bi-temporal (bilateral) electrode placement	Right unilateral electrode placement	Bi-frontal electrode placement
• Electrodes are placed over both temples. The midpoint of each electrode is placed 3 cm above the midpoint of an imaginary line drawn between the external	• The temporal electrode is placed over the right temporal fossa, with the centre of the electrode 2.5 centimetres above the midpoint of a line drawn between the	• The anterior EEG recording electrodes should be moved medially to approximately one centimetre either side of the midline to allow room

canthus of the eye and the tragus of the ear on the left and right temples. This provides the most effective treatment response, however, has the most side effects including disorientation	tragus and the outer canthus of the eye. The centre of the second electrode is placed slightly (one centimetre) to the right of the vertex, which is at the intersection of the line drawn between the nasion and the inion (occipital process), and the line drawn between the tragus of each ear - For left unilateral ECT, the same positions are used, but on the left side of the head. - This method is the least often used, it is the least effective but has the least associated cognitive side effects.	for placement of the treating electrodes. - The midpoint of each treating electrode is placed five centimetres above the outer canthus of each eye, in a parasagittal plane - Concave metal electrodes or adherent disposable electrodes must be used. - This method is the most often used. It is moderately effective with moderate side effects
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- It is recommended that the correct site be identified by measuring the distance above the outer canthus rather than estimating the distance
- Apply a thin vertical line (2-3 drops) of Pretac Conductive Tens Skin Preparation (see Figure 4) then spread laterally with finger for 10cm up to and including the hairline so just feels tacky, covering the area where the thymapad stimulus electrodes will be placed.

Figure 4: Pretac



4.2. Measuring impedance



Figure 5: Thymapad cables

- After placement of the thymapads and treatment cables the Registrar will press the Impedance button on the Thymatron IV™ machine for at least 2 seconds to measure static impedance (see Figure 6).



Figure 6: Impedance test



Figure 7: Conductivity gel

- The impedance reading should be less than 3000, ideally under 2000. The lower the reading the better the conductivity.
- If the impedance reading is greater than 3000 the machine will not treat
- Impedance greater than 3000 indicates there is high resistance and poor contact. To proceed check the connections, common issues are:
 - Insufficient pressure applying the electrode – press more firmly and recheck impedance
 - Placement of thymapad over hair – gently lift the outer edge of the electrode and apply a small amount of Lectron II Conductivity gel (figure 7) between the electrode and skin and push firmly with the palms of the hands on both electrodes and recheck impedance

- If the reading is between 2900 and 3000 the registrar will apply the ECT hand paddles (unattached) to both electrodes during ECT to provide better contact and the Consultant will then be responsible for pressing the treatment button.
- If still unable to progress consult further with the supervising Consultant Psychiatrist
- When the impedance button is pressed the Thymatron IV™ machine starts to collect baseline data and will show 'Ready' when this is complete.
- If there is insufficient information it will show 'No baseline', at this point re-press Impedance button for 2 seconds to gather more information. Repeat as required.

4.3. Monitoring

- The registrar will obtain a baseline (EEG) printout by pressing the **black** start button to the top left of the printout on the Thymatron IV™. Press the **black** start button again to stop the printout
- Check the EEG readout to ensure there are no 'speed bumps' or thick tracing on the EEG.
- If either is present, check the positioning of the 5 recording electrodes – in particular the electrodes behind the ear and on forehead in case of inadequate attachment. If attachment appears adequate recheck the tracing. If still present peel back half of the EEG electrode and add a small amount of electrode gel.
- If the readout is still suboptimal switch the Thymatron IV™ off and on, and commence again cleaning and drying monitoring electrode placement sites and using a new set of monitoring electrodes
- Check there is a good QRS complex (at least 6mm) on the ECG. If less, reposition the left chest lead to obtain greater amplitude
- When the readings are satisfactory the Registrar will inform the Anaesthetist, and confirm it is time to commence anaesthetic for ECT.

4.4. Treatment Administration

An onsite Consultant Psychiatrist who is appropriately credentialed will supervise the Psychiatric Registrar in the administration of ECT.

Registrars will be provided with in house training and an orientation program prior to working in ECT.

- Perform ECT Treatment Team Time Out when the Anaesthetist and ECT Coordinator confirm the anaesthetic and muscle relaxant are working (by absence of fasciculation and/or elapsing of 2 minutes post injection of muscle relaxant) and the mouth guard is in place.

- The registrar will confirm readiness to proceed by stating “ECT treatment will now be delivered”
- Following verbal confirmation that the team is ready the registrar will state “Treating” and press and hold the yellow treatment button until treatment sound has finished
- The registrar will monitor and document the quality of the fit by monitoring both the visual seizure and EEG seizure continuously until EEG seizure is complete plus 15-20 seconds or a drop-in heart rate has stabilised.
- The registrar will inform the Anaesthetist treatment is finished
- The tracing will have a line along the top indicating when the seizure was recorded and should have stopped for at least 3 seconds for outcome calculations to occur
- Analyse the recording and enter results on the Integrated ECT/Anaesthesia record. The Registrar will document the next dose in Consultation with the Treating Consultant, the quality of the fit and comments as needed.

5. Compliance/Performance Monitoring

The FHMHS will maintain a record of trainee experience in the administration of ECT and of Consultant Psychiatrist administration and supervision of ECT. Reports will be monitored by the Head of Service and ECT Working Group annually.

All clinical incidents related to ECT will be reported via the Datix clinical Incident Management System (CIMS) and trended incidents reported to Service 5 Safety Quality Risk Committee.

6. Related Policy Documents

FSFHG:

- Electroconvulsive therapy (FSFH-MENH-POL-0014)
- ECT Coordinators role (FSFH-MENH-POL-0013)
- Electroconvulsive Therapy -Recovery and Discharge (FSFH-MENH-POL-0003)

7. Related Documents

1. [Royal Australian New Zealand College of Psychiatrists \(RANZCP\) Professional Practice Guidelines for the Administration of electroconvulsive therapy \(ECT\) 2019](#)
2. [RANZCP Position statement 74. Electroconvulsive Therapy](#)

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3. [Chief Psychiatrist's Guidelines for the use of Electroconvulsive Therapy in Western Australia 2024](#)
4. [Chief Psychiatrists Standards for the Administration of Electroconvulsive therapy \(ECT\) in WA.](#)

8. Related Legislation

[The Western Australian Mental Health Act 2014](#)

9. Related Standards

NSQHS Standards:

- Clinical Governance
- Comprehensive care
- Recognising and responding to acute deterioration

Chief Psychiatrists standards for clinical care:

- Care planning
- Consumer and carer involvement in individual care
- Physical health care of Mental Health Consumers

10. Authorisation

EXECUTIVE SPONSOR: Service Director, Service 5					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	04/2010	CNS, CI	FH MHS Clinical Governance Committee	FH MHS Clinical Governance Committee	04/2016
2	04/2016	CNS, CI	Service 5 SQ&R	FSFHG Policy Committee	04/2019
3	10/2019	ECT Coordinator/CNS, CI	Service 5 SQ&R	FSFHG Policy Committee	10/2022
4	02/2023	ECT Coordinator/CNS, CI	FH ECT User Group	FSFHG Policy Committee	02/2026
5	08/2025	Snr Policy Coordinator	Minor Update, No change to review date		02/2026